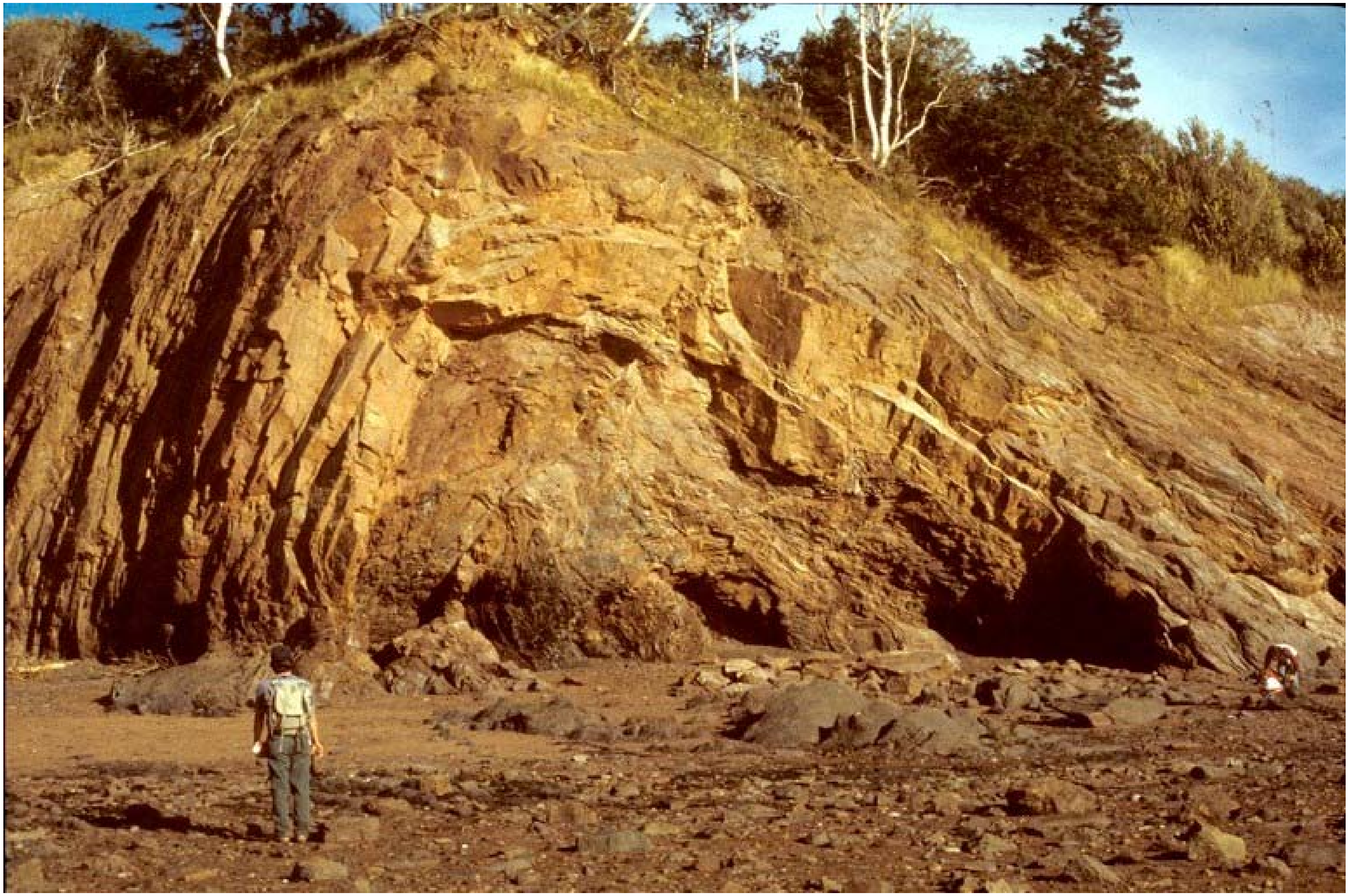
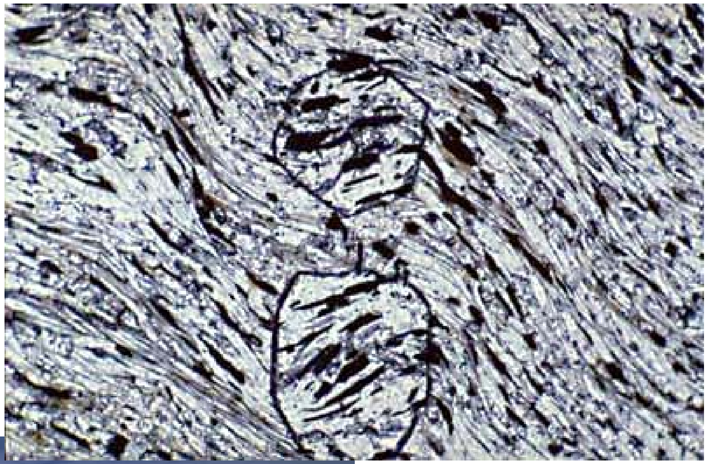
Structural Geology
Fold

FOLDS



Folds

Folds are wave like structures that produced by deformation of bedding, foliation or other planar surfaces in the rocks. They occur on all scales form microscopic to kilometers sizes. They form in all deformational environments from near surface brittle to lower-crust ductile and from simple shear to pure shear. They occur singly and in extensive fold trains



Importance of folding

- ***Hydrocarbon traps.***
- ***Concentration of valuable minerals (saddle-reef deposits) sulfide minerals localized in the hinges of the fold***

Scale types of Folds

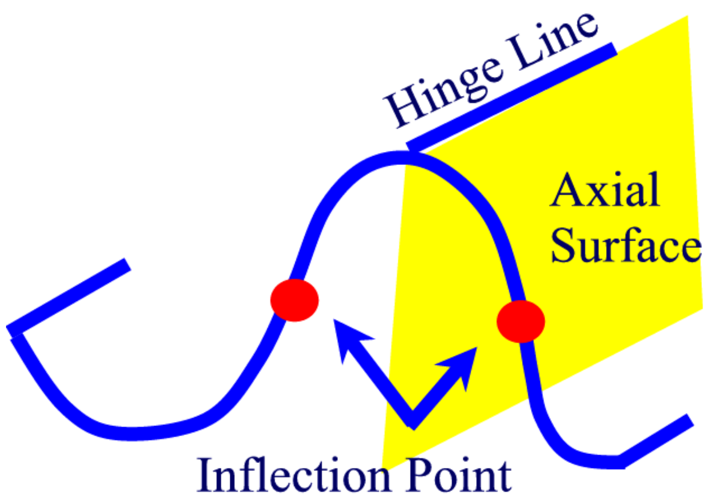
Folds can present in all scales

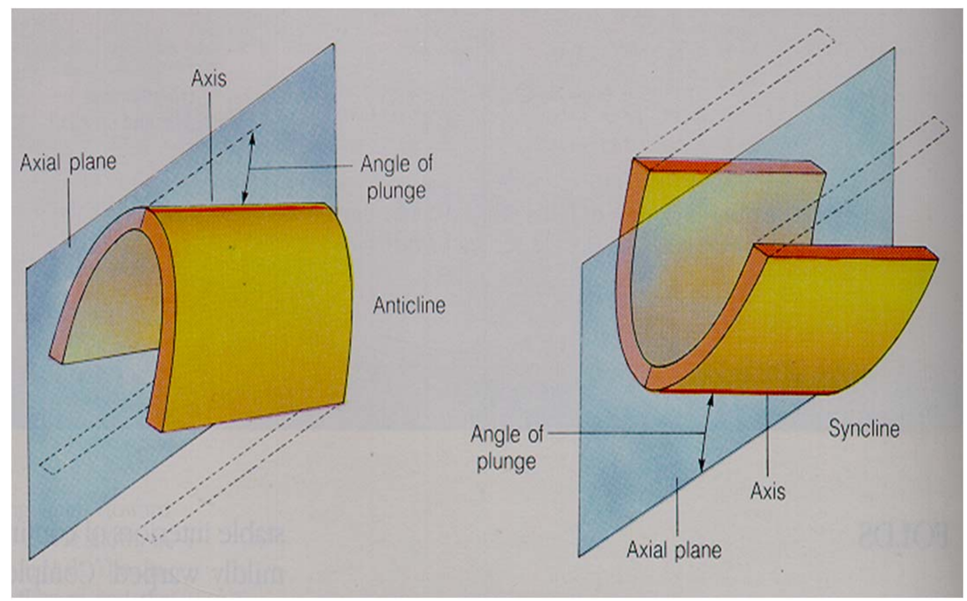
- ***microscopic*** **(require magnification)**
- ***mesoscopic*** **(specimen and outcrop size)**
- ***macroscopic*** **(larger scale)**

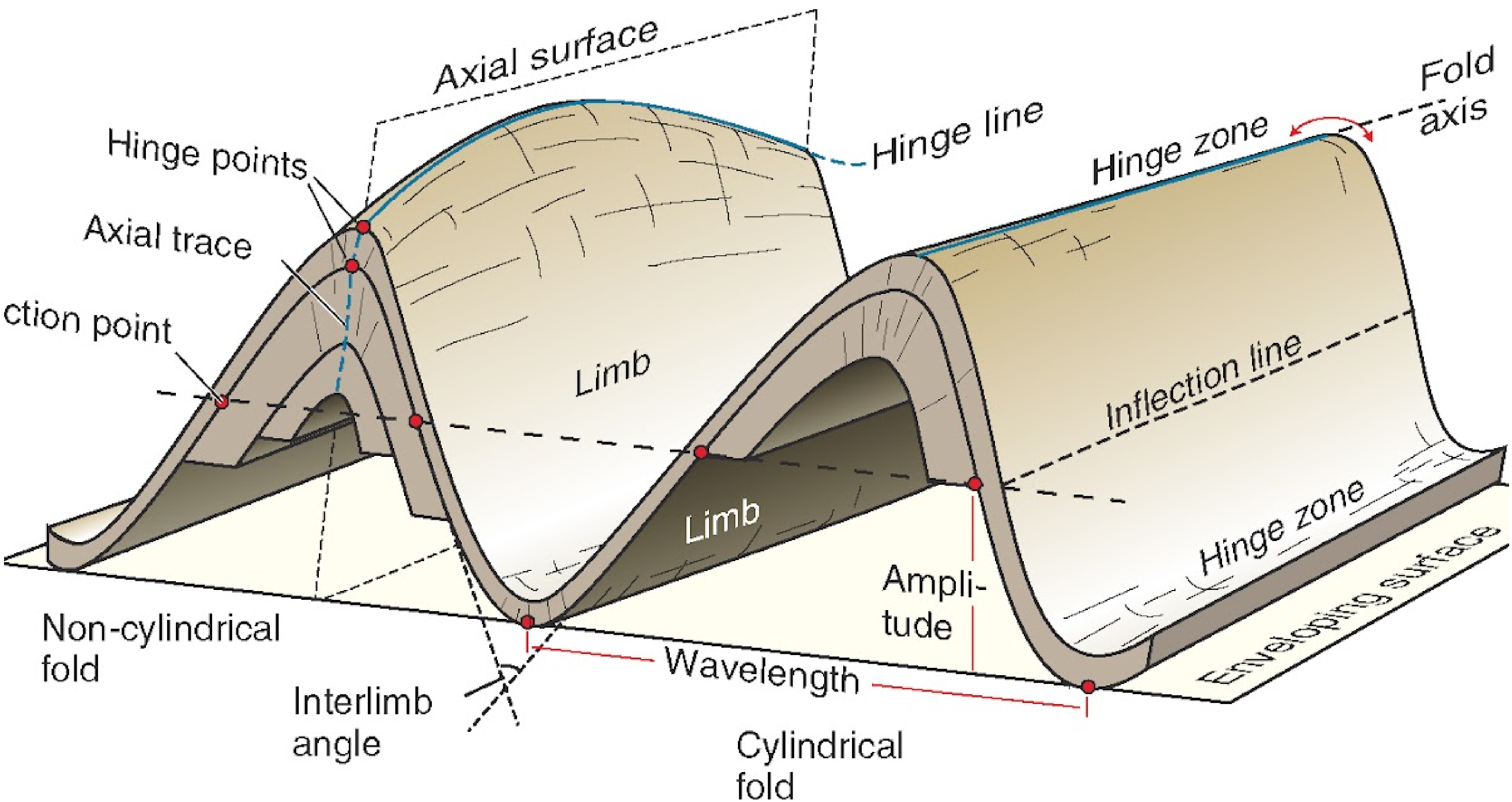
Pumpelly's rule: ***small-scale structures are generally mimic larger-scale.***

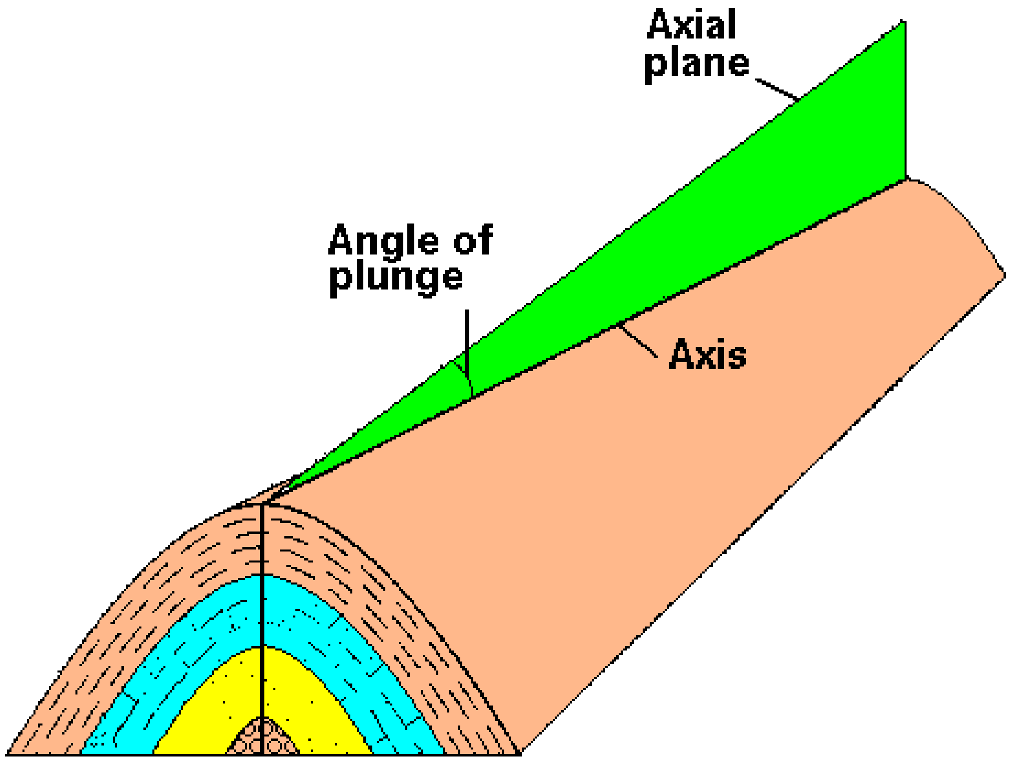
ANATOMY OF FOLDS

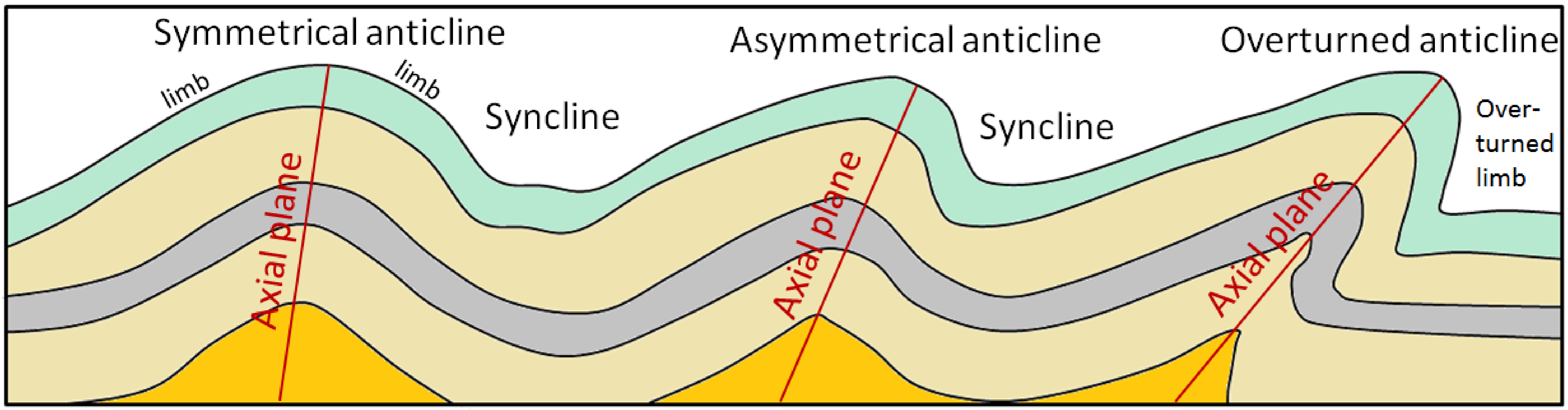
- **Crest, trough, Limbs, hinge zones, fold axis, axial plane, axial surface, plunge, wavelength, inflection point and vergence.**



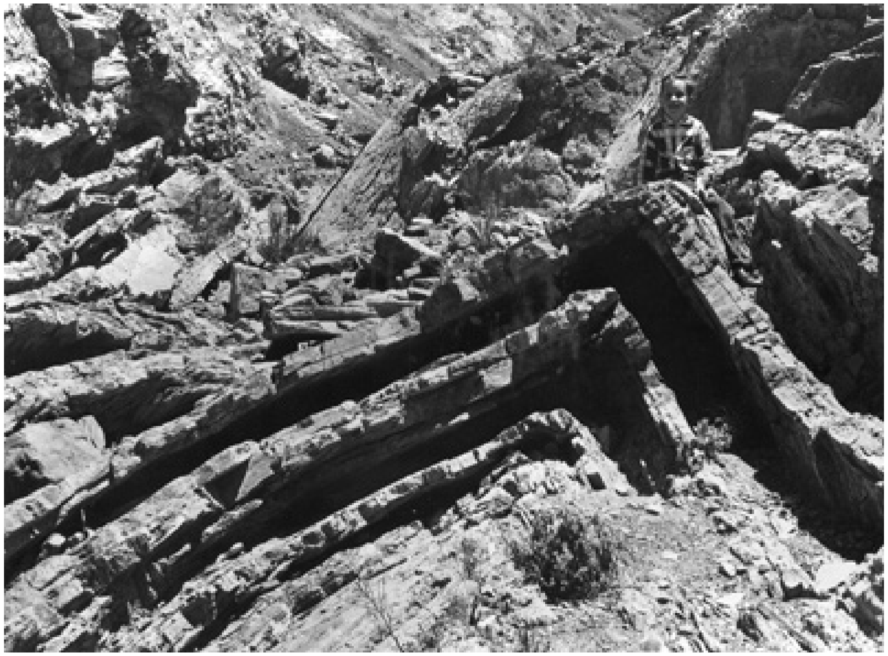



Axial surface
Hinge line
Fold axis
Hinge zone
Hinge points
Axial trace
ction point
Limb
Limb
Inflection line
Hinge zone
Enveloping surface
Amplitude
Non-cylindrical fold
Wavelength
Interlimb angle
Cylindrical fold


Axial
plane
Angle of
plunge
Axis


Symmetrical anticline
limb
limb
Syncline
Asymmetrical anticline
Syncline
Overturned anticline
Over-
turned
limb
Axial plane
Axial plane
Axial plane

Vergence



- **Vergence** of a fold applies only to folds having **one limb that dips more steeply and is shorter than the other-an asymmetric fold**. In symmetrical folds vergence is not a property. **However, small folds on the limbs of symmetrical fold may exhibit vergence**.
- ***Study of vergence may be useful in working out the overall direction of tectonic transport of all structures in an area and help to fix an observer's location on large fold.***

Slip lines: *lines of fibers or slicken-sides on a layer surface that indicate the direction of motion of one layer past another*

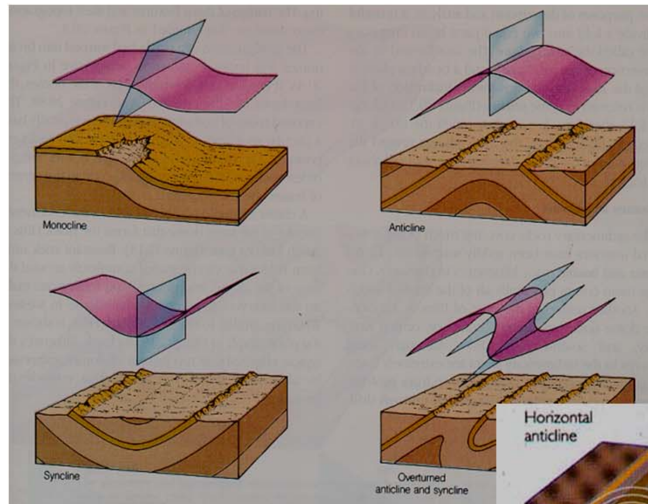
Fold orders

The largest folds in a given area are often called first-order folds, smaller folds on the limbs (flanks) are second order folds.

To relate the geometry of small-to large scale folds enveloping surface is used. The enveloping surface can be constructed through connecting the inflection points. ***Enveloping surfaces are useful for studying folds at outcrop scale or in cross section where many small folds occur on limbs of larger folds, but the geometry of the larger folds not clear.***

Types of Folds

- ***Anticline:*** **folds that are concave towards the older rocks.**
- ***Syncline:*** **folds that are concave towards the younger rocks.**
- ***Antiform:*** **fold is concave downward and rocks may not be older in the middle or age of the rocks is not known.**
- ***Synform:*** **fold is concave upward and rocks in the middle may not be younger or age is not known.**
- ***Dome:*** **layering dips in all directions away from a center point.**
- ***Basin:*** **layering dips inward toward a central point.**
- ***Antiformal syncline:*** **Downward facing syncline in which layering dips away from axis, but the rocks in the center are younger.**
- ***Synformal anticline:*** **upward facing anticline, where in layering dips inward as syncline but the rocks in the center are older.**



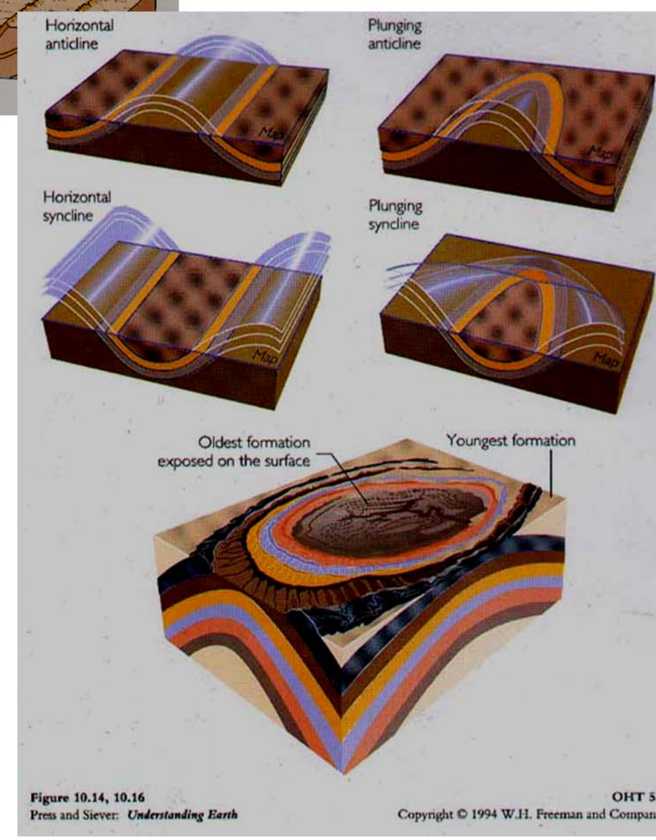


Figure 10.14, 10.16
Press and Siever: *Understanding Earth*
OHT 53

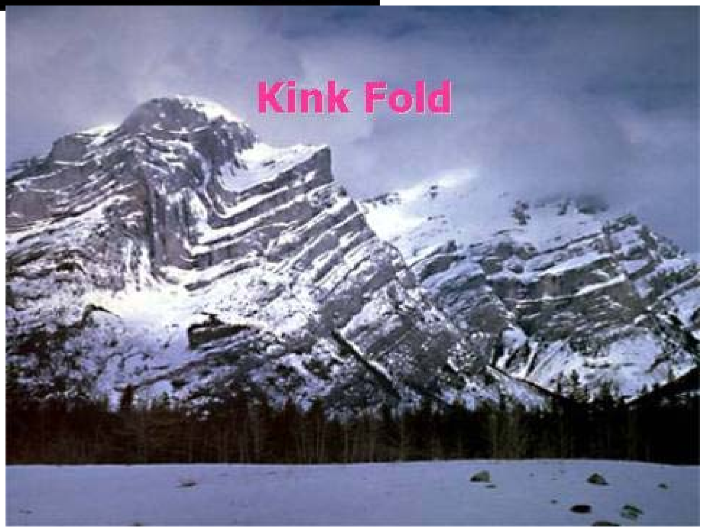
Types of Folds

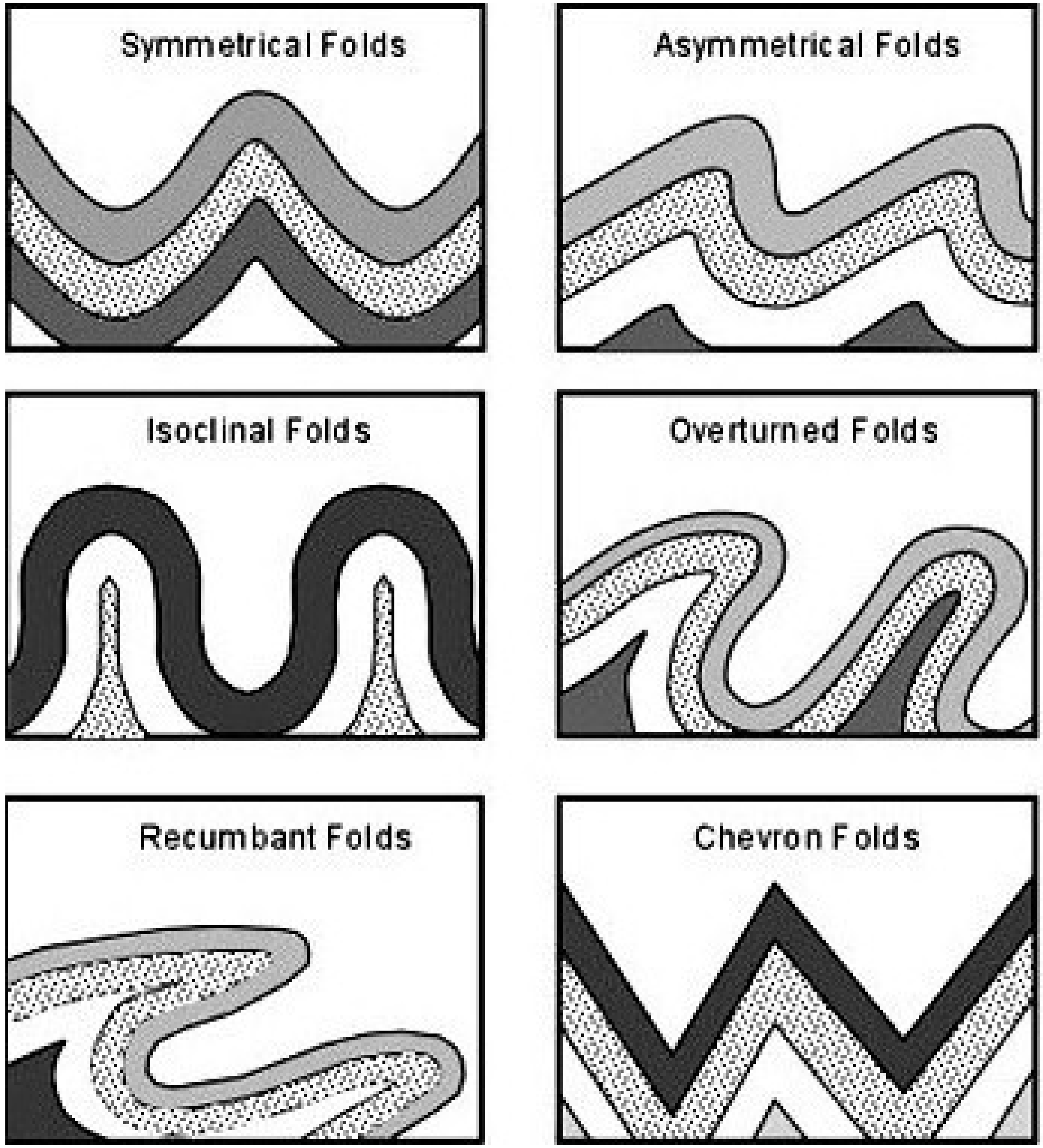
- ***Homocline:*** **rocks that dip uniformly in one direction**
- ***Monocline*****: a local steepening with homocline**
- ***Structural terrace:*** **local flattening of a uniform regional dip**
- ***Cylindrical:*** **The hinges are parallel every where and the fold can be generated by moving the fold axis parallel to itself**
- ***Non-cylindrical:*** **The hinges are not parallel and can converge in one point**
- ***Sheath folds:*** **are non-cylindrical and closed at one end the fold hinges curve within axial surface**
- ***Upright folds:*** **have vertical axial surface**
  Overturned folds**: have one inverted limb**
- ***Reclined folds:*** **axes plunge at nearly same angle as the dip of the axial surface, plunge of the axis normal or at high angle to the strike of the axial plane**
- ***Recumbent folds:*** **Have horizontal axes and axial surfaces.**
- ***Isoclinal folds:*** **are tight folds wherein axial surfaces and limbs are parallel**

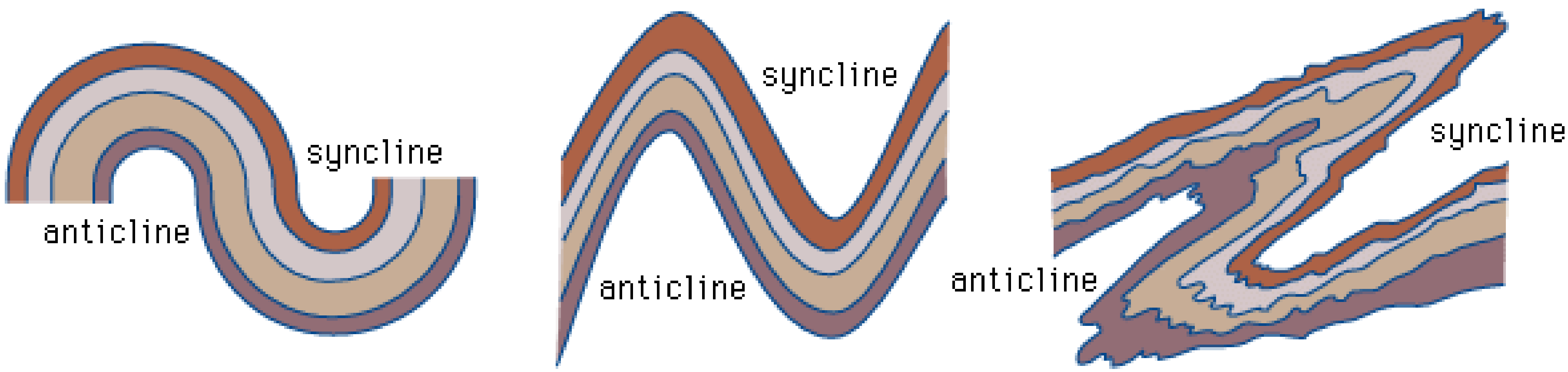
To distinguish between the different type of folds Fig. 14.13 (after Fleuty 1964) is used.

Classification of folds based on the bedding thickness, and hinge curvature

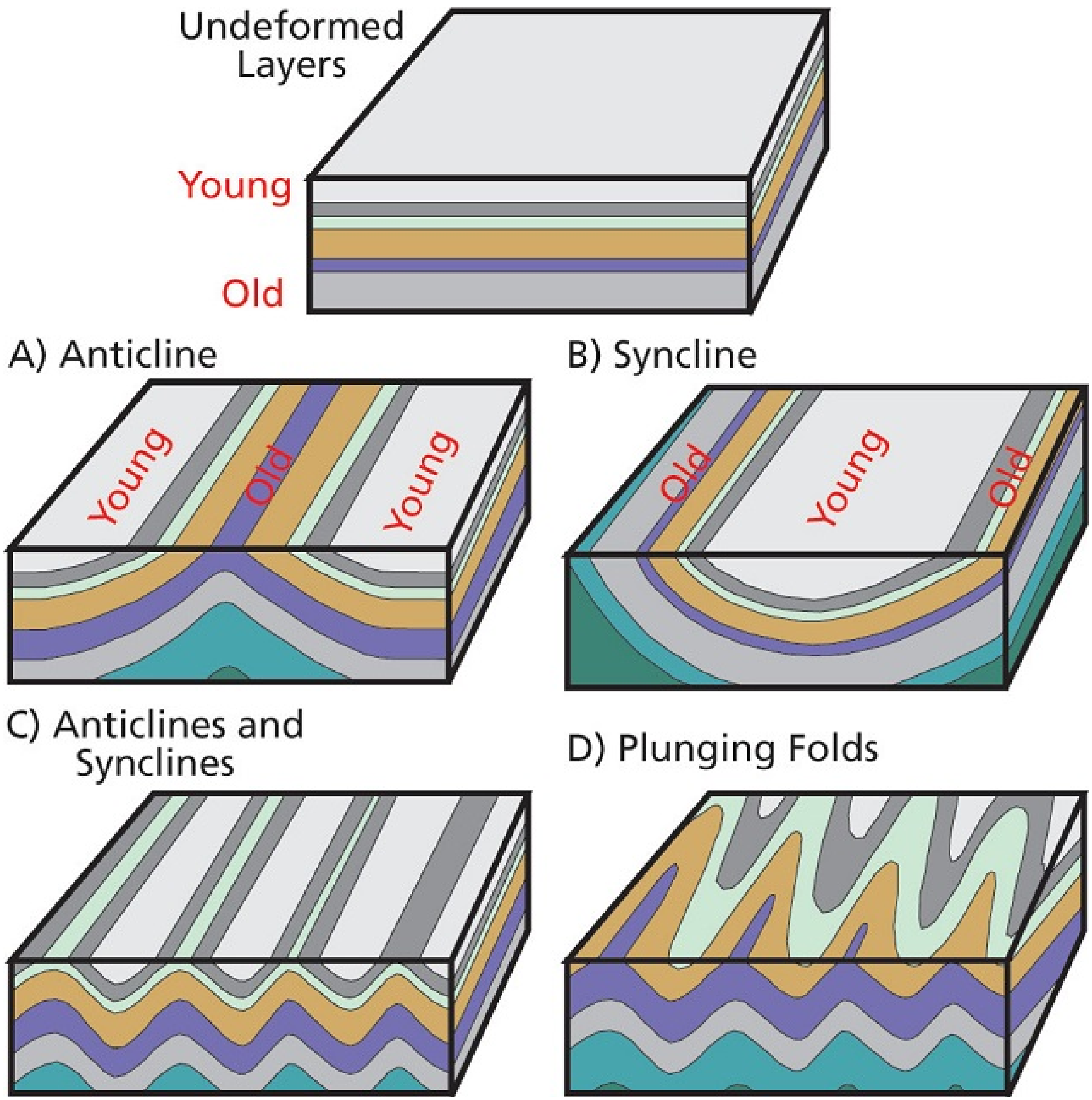
- ***Parallel folds:*** *folds maintain constant thickness*
- ***Concentric folds:*** *parallel folds in which folded surfaces define circular arcs and maintain the same center of curvature.*
- ***Ptygmatic folds:*** *nearly concentric shape, attenuated limbs and intestinal appearance.*
- ***Similar folds:*** *maintain the same shape throughout a section but not necessarily with the same thickness.*
- ***Chevron and kink folds:*** *have sharp angular hinges and straight limbs.*
- ***Disharmonic:*** *shape or wavelength changes from one layer to another.*
- ***Supratenuous folds:*** *synclines are thickened and anticlines are thinned. These folds are usually non-tectonic form in unconsolidated sediments and when uplift is taking place.*
- ***Fault-bend and fault-propagation folds:*** *these type of folds associated with thrust fault*




Symmetrical Folds
Asymmetrical Folds
Isoclinal Folds
Overturned Folds
Recumbant Folds
Chevron Folds

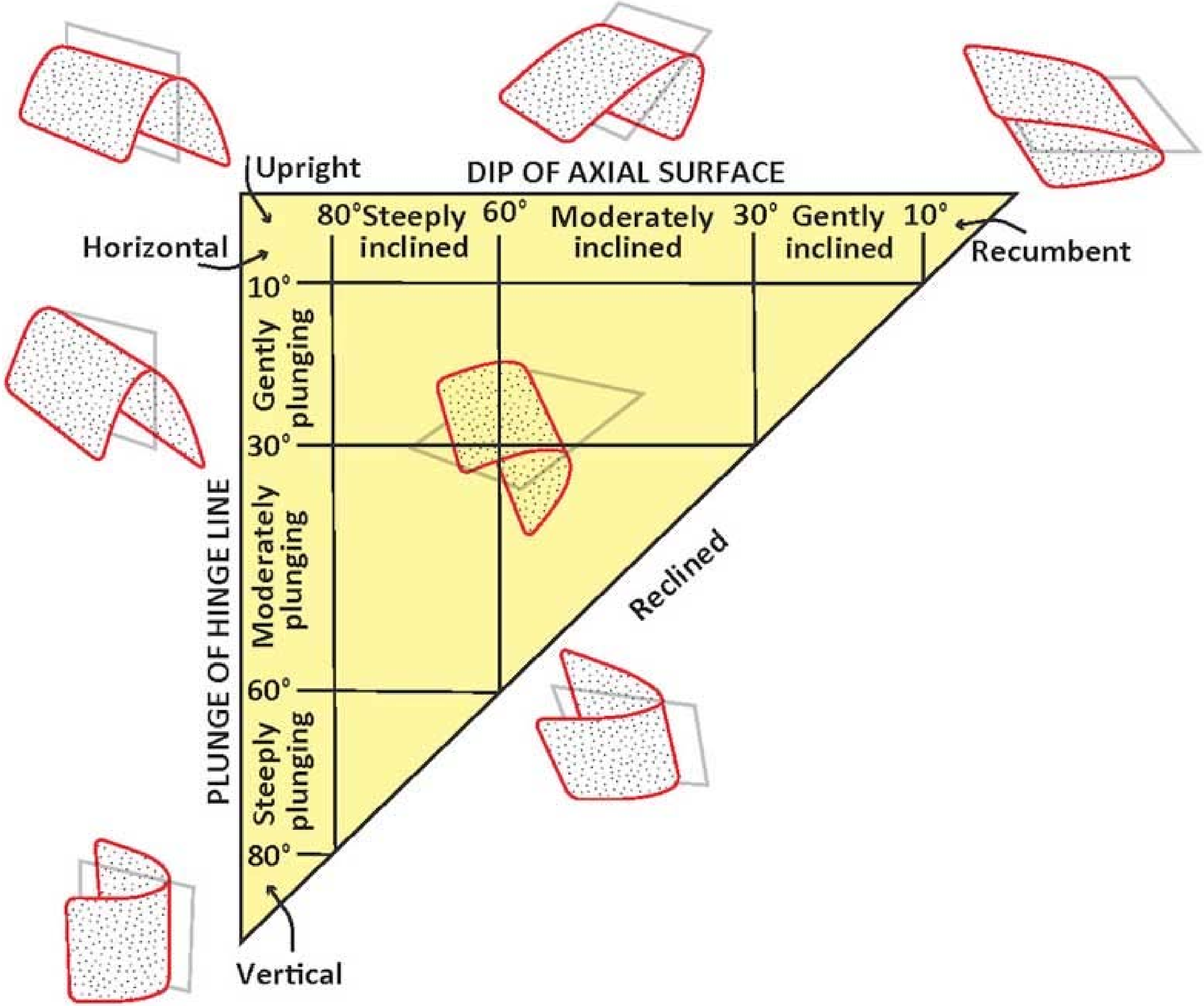


parallel fold　similar fold　flow fold

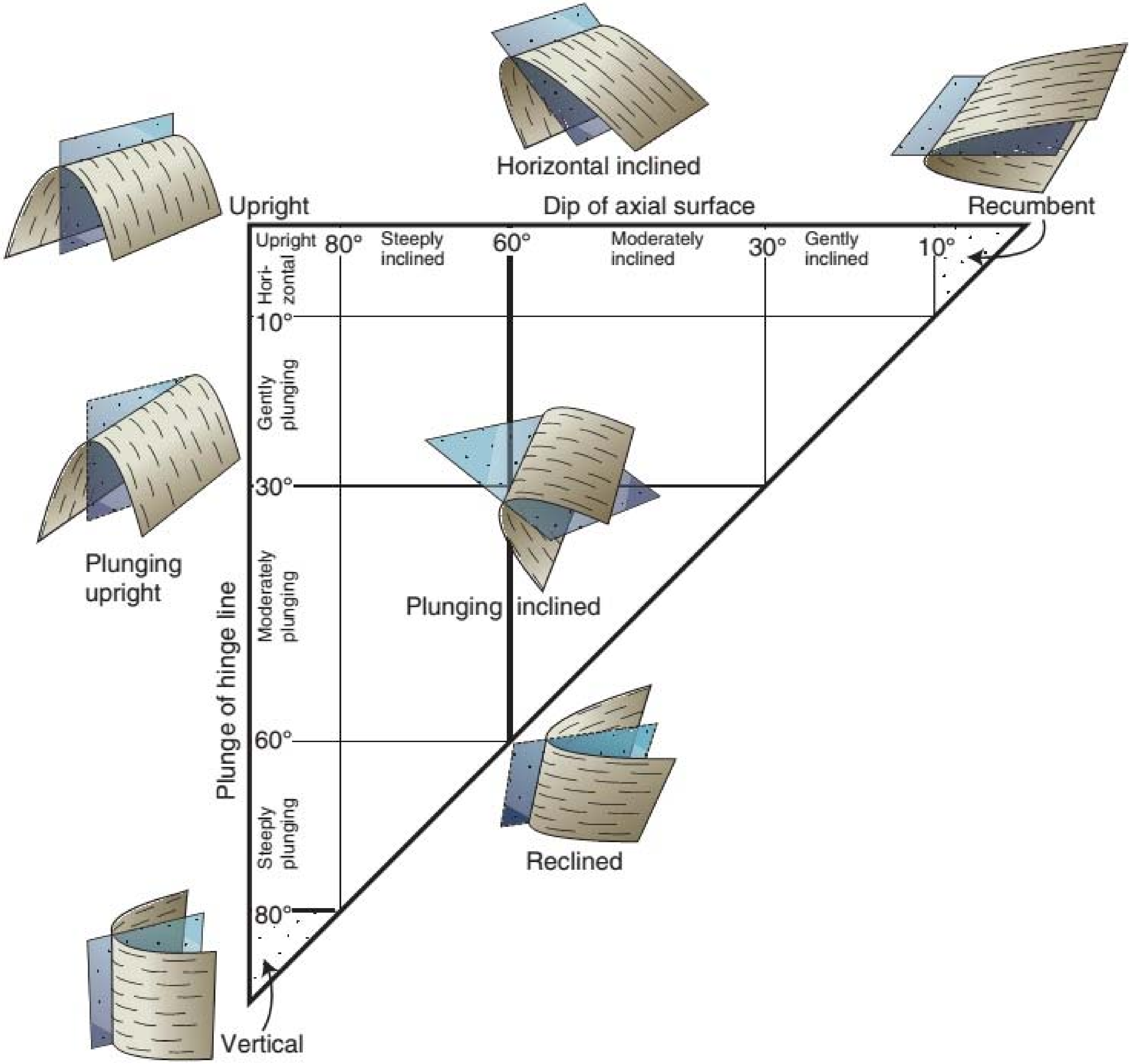

Undeformed Layers
Young
Old
A) Anticline
Young
Old
Young
B) Syncline
Old
Young
Old
C) Anticlines and Synclines
D) Plunging Folds

Parasitic folds are used to determine the position in a fold

- *parasitic or small size fold on the limb of big size fold can be used to determine the position as they have Z sense of rotation clockwise in one limb and S sense of movement anti-clockwise in the opposite limb. W and M sense of movement are found at the hinge of the big size fold.*
- *Stereonet is also used to determine the direction, vergence, and sense of movement of big fold by plotting the vergence and parasitic small folds.*



Upright
DIP OF AXIAL SURFACE
Horizontal
80° Steeply inclined
60°
Moderately inclined
30°
Gently inclined
10°
Recumbent
10°
Gently plunging
30°
Moderately plunging
60°
Steeply plunging
80°
Vertical
PLUNGE OF HINGE LINE
Reclined


Horizontal inclined
Upright
Dip of axial surface
Recumbent
Upright
80°
Steeply inclined
60°
Moderately inclined
30°
Gently inclined
10°
Hori- zontal
10°
Gently plunging
30°
Plunging upright
Plunging inclined
Plunge of hinge line
Moderately plunging
60°
Reclined
Steeply plunging
80°
Vertical

FOLDS CLASSIFICATION

- **<u>Fleuty Classification</u>*:***

based on interlimb angle and hinge area

Gentle, Open, Closed, Tight, Isoclinal and Elastica

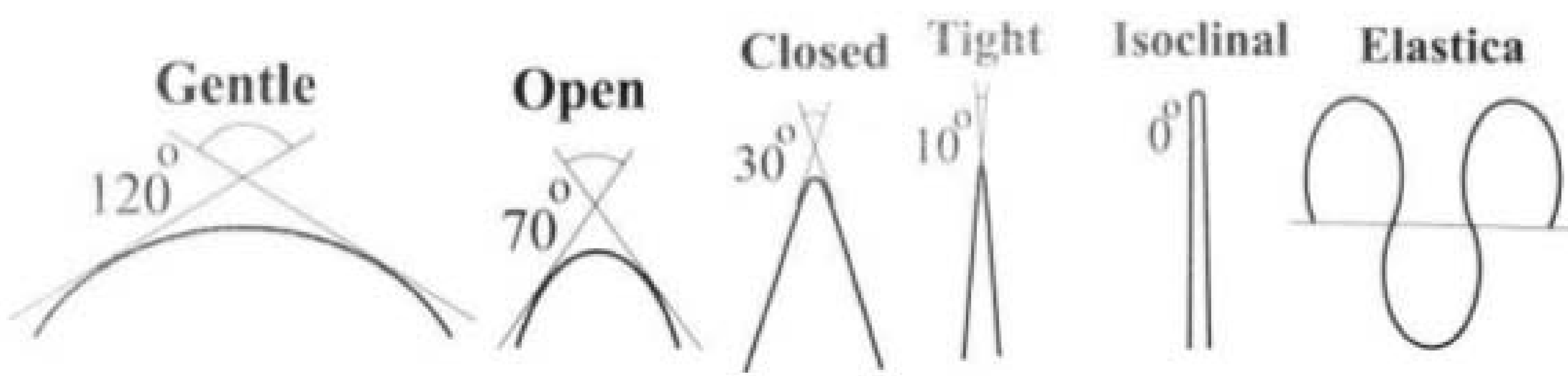
Fleuty's classification of folds

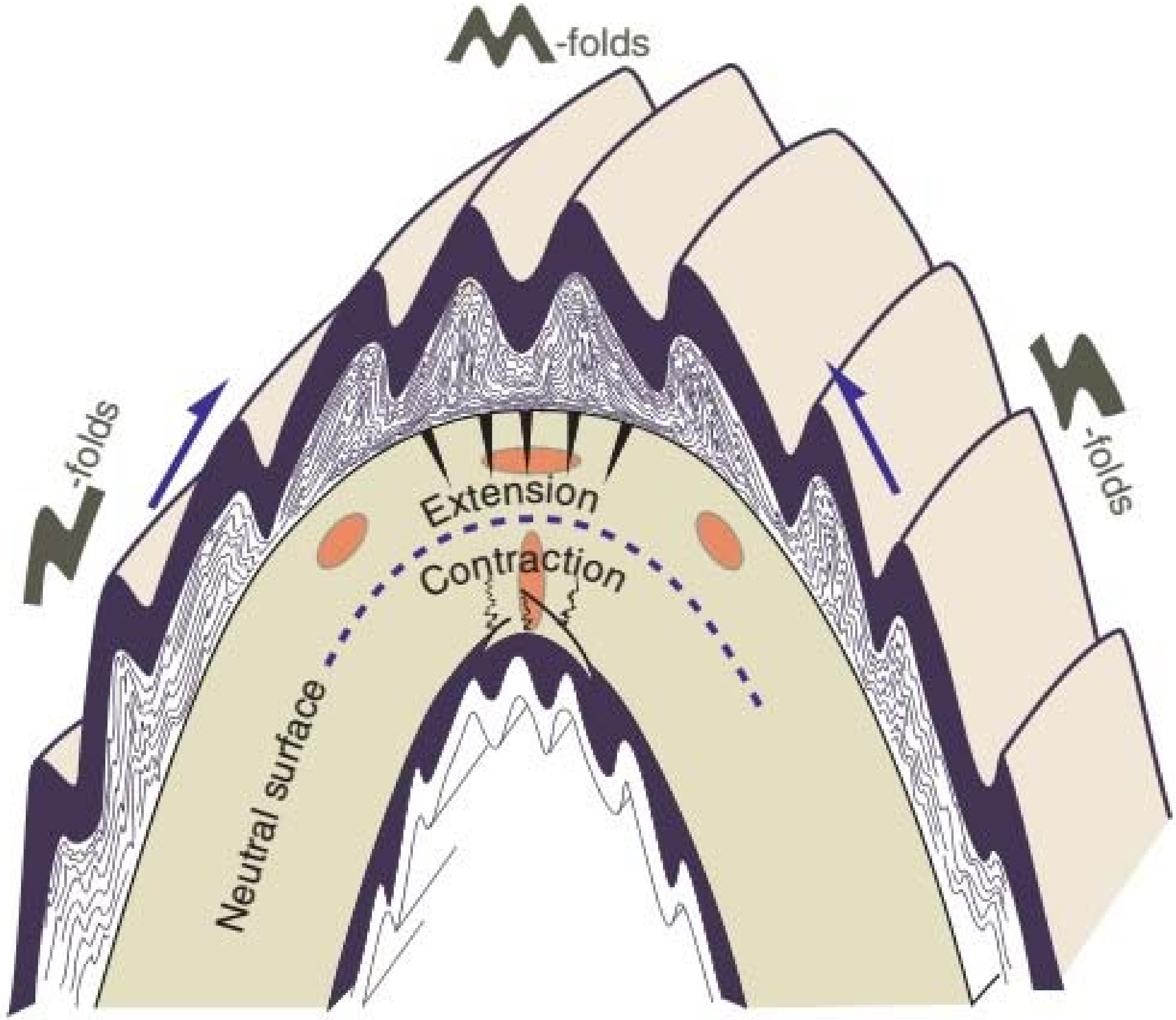
180^o	$>$	θ	$>$	120^o	gentle
120^o	$>$	θ	$>$	70^o	open
70^o	$>$	θ	$>$	30^o	close
30^o	$>$	θ	$>$	0^o	tight
0^o		θ	$=$		isoclinal
θ	$<$	0^o			mushroom

7. Fluety's classification on the basis of interlimb angle

Given by Fluety (1964)

- **Gentle fold -** interlimb angle between 180-120°
- **Open fold -** interlimb angle between 120-70°
- **Close fold -** interlimb angle between 70-30°
- **Tight fold -** interlimb angle < 30° but > 0°
- **Isoclinal fold –** with subparallel limb.
- **Fan Fold –** with negative interlimb angle.





-folds
-folds
-folds
Extension
Contraction
Neutral surface

Fold Classification – Ramsay 1967

Start with a profile plane view of a fold (constructed by rotation, or photographed in the field looking downplunge). .1

Mark the hinge points and inflection points on the two bounding surfaces of the folded layer **.2**

Draw the tangents to the folded layer at the hinge points. This is the zero dip (α **= 0**) reference .3

At α = 0, measure the orthogonal hinge thickness t_o **.4**

Construct other tangents at other α angles .5

Measure the orthogonal thickness (t_α) between these tangents for these α angles **.6**

Determine the ratio: $t_\alpha' = t_\alpha / t_o$ .7

Plot t_α' as ordinate against α as abscissa **.8**

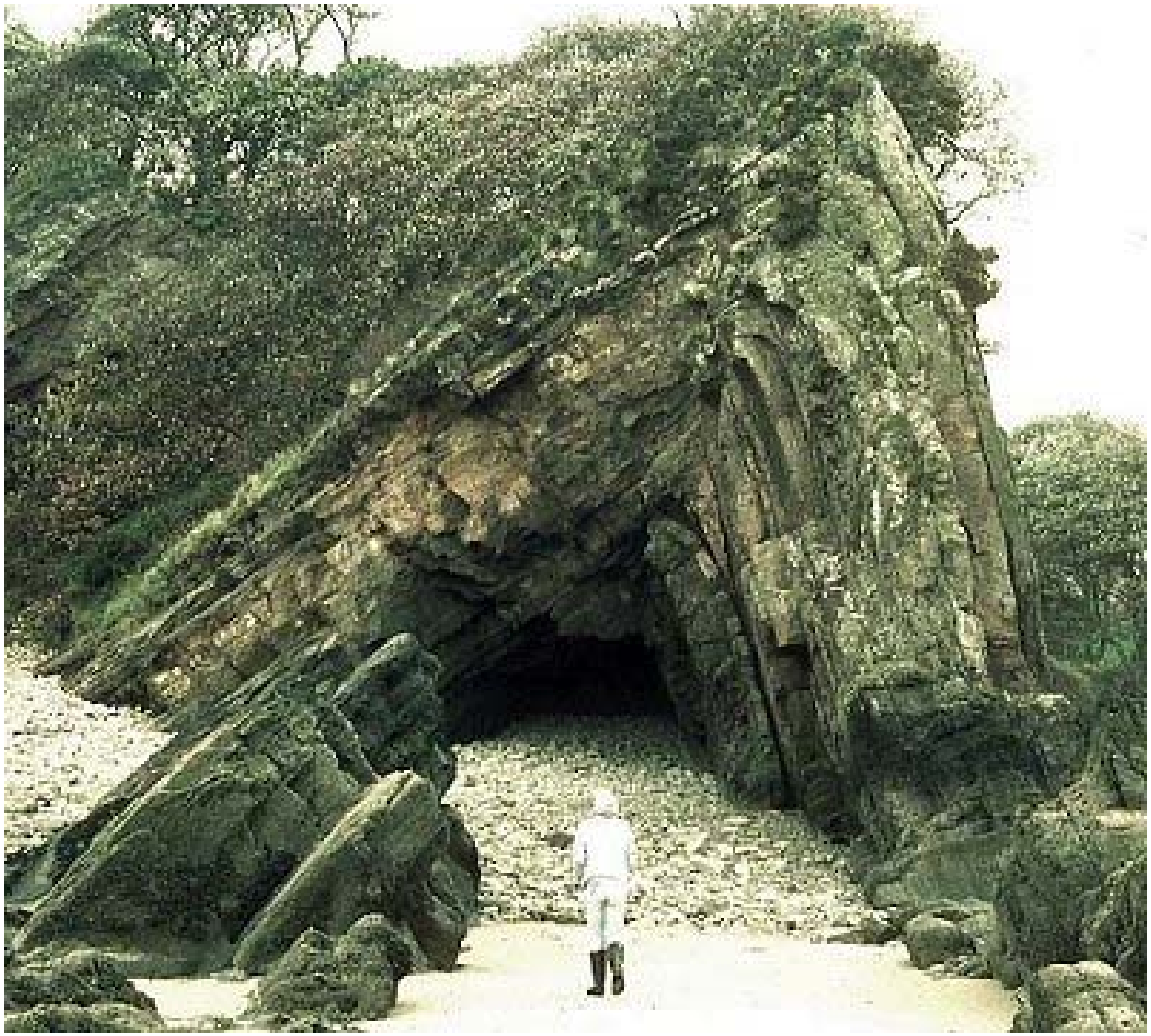
Repeat for all values of α .9

Fold Classification - **Parallel Folds**

- Normal thickness (**t**), perpendicular to the layer, is constant throughout the fold, i.e., $t_{\alpha}' = t_{\alpha} / t_{o} = 1$
- Some parallel fold are concentric; i.e., have constant, circular curvature
- Parallel folds are typical of competent layers

- Layer thickness, measured parallel to the axial plane, is greater on the limbs (**T**) than that around the hinge ($\mathbf{T_o}$), i.e., ($\mathbf{T > T_o}$)

Chevron fold



Similar Folds

- Have variation in their layer thickness, **t**
- **The orthogonal layer thickness reduces on the limb**
- The orthogonal thickness varies as: $\mathbf{t_\alpha' = cos}\ \alpha$
- The curvature of the bounding surfaces are identical (hence the word ‘similar’)
- **The layer thickness measured parallel to the axial plane (T) is constant ($T_\alpha = T_o = t_o$)**

Similar fold

Intermediate Fold Styles – Ramsay 1967

- **The similar and parallel folds are not the end members of fold style. Other styles fall outside of these two**

- Progressive deformation (e.g., flattening) may change one geometry to another

- **Ramsay proposed analyzing the variation of the layer thickness with the angle of dip within a quarter fold wave sector**

Dip Isogons

Lines joining points of equal dip (normal to tangents)

Drawn at different α angles (at 10^o intervals)

Isogons can be **parallel, converging, or diverging**

The *sense is from the outer arc to the inner arc*

Parallel isogons:

The average inner and outer curvatures are equal

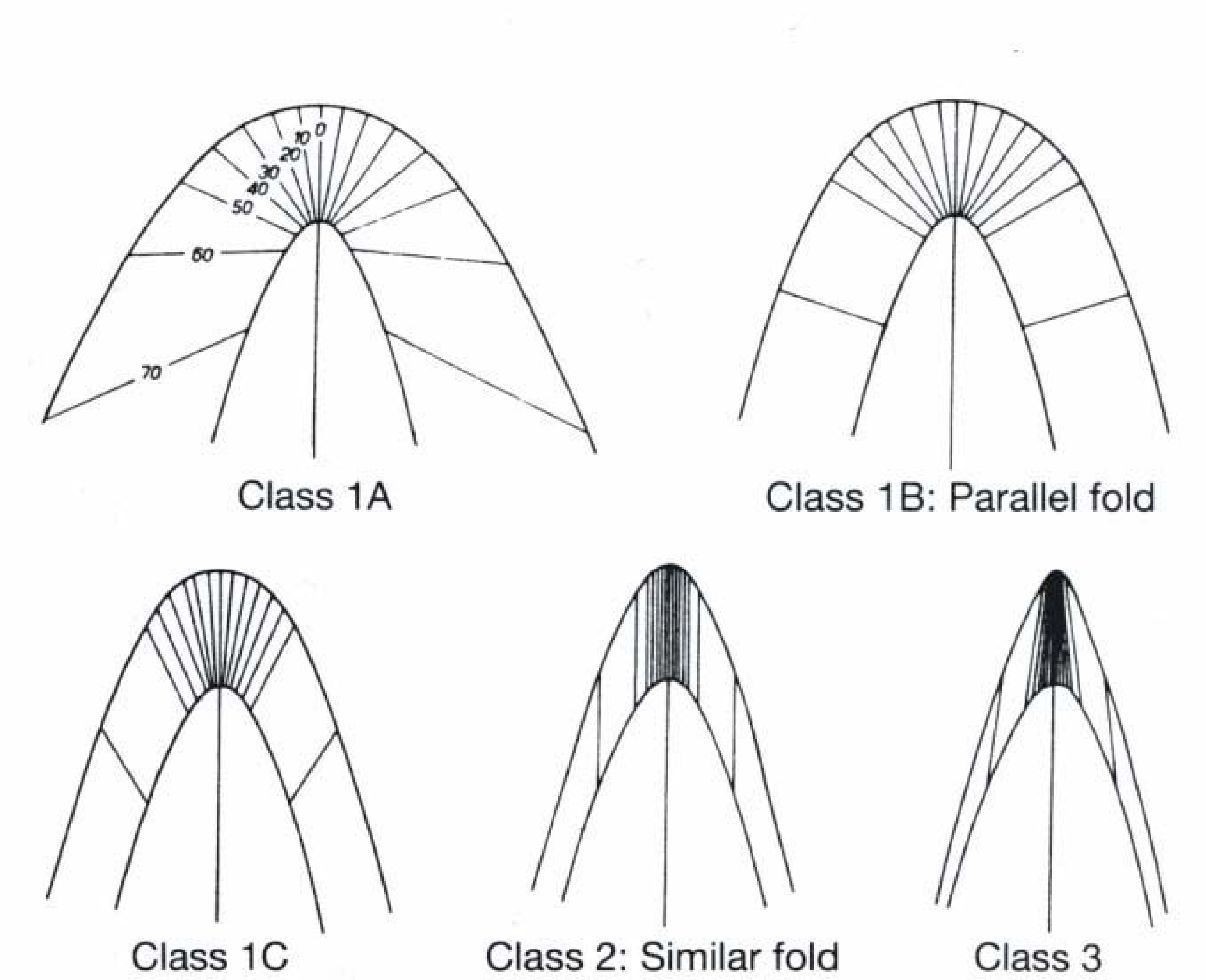
Converging isogons:

Inner arc curvature exceeds that of outer arc

Divergent isogons:

Outer arc curvature exceeds that of the inner arc.

Dip isogons Twiss andMoores, 1992



Fold Classification …

- On the t_α' against α diagram:
- **All other folds fall on either side of the two lines:**

  t_α' = 1 (parallel fold = class 1B)

  t_α' = cos α (similar fold = class 2)

- Class 1A folds lie above class 1B parallel folds.
- **Class 1A folds are thicker on the limb than near the hinge**
- Class 1C, class 2, and class 3 folds all show thinning on the limb

Fold Classes

Class	Isogon	Subclass	Isogon
1	Convergent	1A	Strongly convergent
		1B - Parallel fold	Normal to layers
		1C	Weakly convergent
2	Parallel	Similar fold	
3	Divergent		